

UPAS - Underground Protection Analysis System

Complete Project Specification Package for Code Agent

Version 1.0

Project Overview

UPAS is a professional web-based engineering analysis application.

It provides project definition, threat modeling, soil modeling, underground structure modeling, engineering calculations, interactive 3D visualization, review tools and reporting.

The 3D model is the main engineering review interface. Every engineering value must be connected to a visual object.

Technology Stack

Frontend:

- React
- TypeScript
- Vite
- Three.js
- React Three Fiber
- Drei
- Tailwind CSS

Future Backend:

- Node.js
- TypeScript
- PostgreSQL
- REST API

System Architecture

The application must use modular architecture:

UI Layer

Data Models

Engineering Calculation Engine

3D Visualization Engine

Database Layer

Reporting Engine

Rules:

- Keep calculations independent from UI.
- Do not place engineering logic inside React components.
- Every 3D object must have linked engineering data.

Main Modules

1. Project Management:

Create, save, load and export projects.

2. Threat Definition:

Support penetrating scenarios, surface explosion scenarios and custom scenarios.

3. Soil Model:

Support multiple soil layers with thickness, density, strength and properties.

4. Structure Model:

Support box structures, arch structures, circular tunnels and underground facilities.

3D Visualization Specification

The 3D engine must display:

- Ground surface
- Soil layers
- Threat object
- Movement path
- Impact/explosion point
- Blast visualization

- Underground structure
- Dimensions
- Engineering labels
- Loads

User interactions:

- Rotate
- Zoom
- Pan
- Select objects
- Show/hide layers
- Display properties
- Create sections

Engineering Review Mode must show dimensions, thicknesses, materials and status.

Engineering Calculation Engine

The calculation engine must be separated from the interface.

Modules:

- Penetration
- Blast
- Loads
- Structure
- Safety

Input:

Complete project model

Output:

Analysis results.

Database Design

Initial data files:

bombs.json

soils.json

materials.json

structures.json

coefficients.json

The system must avoid hardcoded engineering values.

User Interface

Main screens:

1. Dashboard
2. New Project Wizard
3. 3D Analysis View
4. Results Dashboard
5. Reports

The interface should support Arabic RTL.

Development Roadmap

Phase 1:

- React project setup
- TypeScript configuration
- Folder architecture
- Data models
- Basic 3D scene

Phase 2:

- Terrain
- Soil layers
- Structure models
- Threat visualization
- Dimensions and labels

Phase 3:

- Input forms
- Validation
- Project storage

Phase 4:

- Engineering calculation modules
- Results processing

Phase 5:

- PDF reports
- Export

Phase 6:

- User accounts
- Cloud projects
- CAD/BIM integration

Development Rules

The Code Agent must:

- Build production quality code.
- Use clean architecture.
- Create reusable components.
- Document modules.
- Maintain scalability.
- Complete each phase before moving forward.

First Action Required From Code Agent

Before writing code:

1. Analyze the specification.
2. Create architecture proposal.

3. Create folder structure.
4. Explain implementation plan.
5. Wait for approval.

Do not create a simplified demo.

Build a scalable engineering application.